



Assignment – 2

Schema Tables

1) MEMBER Table

Column Name	Data Type	Description
Member_Id	INT	Unique Member ID
Member_Name	VARCHAR(30)	Name of the Library member
Member_Address	VARCHAR(50)	Address of the member
Acc_Open_Date	DATE	Date of membership
Membership_Type	VARCHAR(20)	Lifetime/Annual/Half Yearly/Quarterly
Fees_Paid	DECIMAL(10,2)	Membership fees paid
Max_Books_Allowed	INT	Total books allowed
Penalty_Amount	DECIMAL(10,2)	Penalty amount due

2) BOOKS Table

Column Name	Data Type	Description
Book_No	INT	Book identification number
Book_Name	VARCHAR(30)	Name of the book
Author_Name	VARCHAR(30)	Author of the book
Cost	DECIMAL(10,2)	Cost of the book
Category	VARCHAR(15)	Science/Fiction/Database/RDBMS/0thers



3) ISSUE Table

Column Name	Data Type	Description
Lib_Issue_Id	INT	Library Issue ID
Book_No	INT	Book number issued
Member_Id	INT	Member who issued the book
Issue_Date	DATE	Date of issue
Return_Date	DATE	Date of return

Questions

Note: Refer to the above schema tables for column names and datatypes.

Q1) Create the tables Member, Books, and Issue WITHOUT constraints.

Q2) Display the structure of all three tables using DESC.

Q3) Drop the Member table.

Q4) Create the Member table again with these constraints:

- Member_Id must be PRIMARY KEY
- Membership_Type must allow only:
('Lifetime', 'Annual', 'Half Yearly', 'Quarterly')

Q5) Alter Member table to increase Member_Name size from 30 to 40.

Q6) Add a column Reference (VARCHAR (30)) to Issue table.

Q7) Drop the column Reference from Issue table.

Q8) Rename the table Issue to Lib_Issue.

Q9) Insert the following records into Member table:

Member ID	Member Name	Member Address	Acc_Open_Date	Membership_type	Fees_Paid	Max_Books_Allowed	Penalty_Amount
1	Richa Sharma	Pune	10-12-05	Lifetime	25000	5	50
2	Garima Sen	Pune	current date	Annual	1000	3	Null



Q10) Insert at least 5 more records with suitable data into Member table.

Q11) Try to decrease Member_Name size from 40 to 20.

If it fails, mention the reason.

Q12) Add a CHECK constraint:

Max_Books_Allowed must be less than 100.

Q13) Create a backup table named MemberBackup using the Member table.

Use the following query:

```
CREATE TABLE MemberBackup AS  
SELECT * FROM Member;
```

Q14) Add named constraints:

- Max_Books_Allowed < 100
- Penalty_Amount <= 1000

Q15) Drop the Books table.

Q16) Create Books table again with constraints:

- Book_No PRIMARY KEY
- Book_Name NOT NULL
- Category must be one of:
('System','Fiction','Database','RDBMS','Others','Science')

Q17) Insert the following data into Books table:

Book_No	Book_Name	Author_Name	Cost	Category
101	Let Us C	Denis Ritchie	450	System
102	Oracle Complete Ref	Loni	550	Database
103	Mastering SQL	Loni	250	Database
104	PL SQL Ref	Scott Urman	750	Database
105	National Geographic	Adis Scott	1000	Science



Q18) Insert at least 5 more records into Books table.

Q19) View all data from Member and Books tables using SELECT.

Q20) Drop Lib_Issue table.

Q21) Create Issue table again with constraints:

- Lib_Issue_Id PRIMARY KEY
- Book_No FOREIGN KEY REFERENCES Books(Book_No)
- Member_Id FOREIGN KEY REFERENCES Member(Member_Id)

Q22) Insert the following records into Issue table:

Lib_Issue_Id	Book_No	Member_Id	Issue_Date
7001	101	1	2024-12-10
7002	102	2	2024-12-25
7003	104	1	2025-01-15
7004	101	1	2025-02-04
7005	104	2	2025-03-15
7006	101	3	2025-04-18

Q23) Try inserting an Issue record with Member_Id not existing in Member table.
Observe and write the error.

Q24) Try deleting Member_Id = 1 from Member table.
Observe and write the foreign key error.

Q25) Update Return_Date for Issue_Id 7004 and 7005 to 15 days after Issue_Date.

Q26) Delete all Issue records where Member_Id = 1 and Issue_Date is before '2006-12-10'.

Q27) Delete all Books where Category is not Database or RDBMS.

Q28) Drop all three tables: Issue, Member, Books.